

Deep Learning Portfolio Examination

Authors: Damian (Matric No. 10012545) Bhuvana Chandra Thota (Matric No. 10001026)

Part - 1: Stabilizing the Existing Codebase

Performance Report

Each of the four recovered diagnostic image profiles (cells, chest, lesions, orgs) was benchmarked against all three architectures (AlexNet, VGG16, ResNet18) using the reconstructed, config-driven pipeline. The complete per-epoch training and validation logs and the final held-out test metrics are reproduced below for every dataset-model permutation.

Dataset Used – cells

Blood cell microscopy (8 classes, RGB) | Expected minimum test accuracy: 90%

1. cells – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.9994 - Train Acc: 61.03% | Val Loss: 0.4850 - Val Acc: 80.03%
Epoch [02/15] | Train Loss: 0.4744 - Train Acc: 82.65% | Val Loss: 1.0726 - Val Acc: 67.08%
Epoch [03/15] | Train Loss: 0.3363 - Train Acc: 88.79% | Val Loss: 2.6263 - Val Acc: 49.96%
Epoch [04/15] | Train Loss: 0.3045 - Train Acc: 90.04% | Val Loss: 0.4279 - Val Acc: 85.59%
Epoch [05/15] | Train Loss: 0.2560 - Train Acc: 91.81% | Val Loss: 1.7035 - Val Acc: 57.72%
Epoch [06/15] | Train Loss: 0.2185 - Train Acc: 92.74% | Val Loss: 4.3006 - Val Acc: 50.84%
Epoch [07/15] | Train Loss: 0.2042 - Train Acc: 93.28% | Val Loss: 2.9192 - Val Acc: 50.77%
Epoch [08/15] | Train Loss: 0.1847 - Train Acc: 94.08% | Val Loss: 0.1911 - Val Acc: 93.78%
Epoch [09/15] | Train Loss: 0.1532 - Train Acc: 95.16% | Val Loss: 0.9190 - Val Acc: 73.88%
Epoch [10/15] | Train Loss: 0.1598 - Train Acc: 95.10% | Val Loss: 0.4752 - Val Acc: 84.56%
Epoch [11/15] | Train Loss: 0.1534 - Train Acc: 95.09% | Val Loss: 0.2879 - Val Acc: 89.90%
Epoch [12/15] | Train Loss: 0.1446 - Train Acc: 95.58% | Val Loss: 1.7349 - Val Acc: 61.23%
Epoch [13/15] | Train Loss: 0.1339 - Train Acc: 95.62% | Val Loss: 0.2969 - Val Acc: 91.95%
Epoch [14/15] | Train Loss: 0.1286 - Train Acc: 95.83% | Val Loss: 0.5742 - Val Acc: 83.03%
Epoch [15/15] | Train Loss: 0.1086 - Train Acc: 96.25% | Val Loss: 0.3102 - Val Acc: 90.56%
Best checkpoint saved: best_model/cells_AlexNet.pth (epoch=8, val_acc=93.78%)
Training Complete!
```

Final Test Results (AlexNet on cells): Accuracy = 95.50% | Precision (macro) = 0.9551 | Recall (macro) = 0.9482 | Macro F1 = 0.9510

2. cells – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
```

```
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.8983 - Train Acc: 66.14% | Val Loss: 0.8445 - Val Acc: 69.79%
Epoch [02/15] | Train Loss: 0.5036 - Train Acc: 82.52% | Val Loss: 4.6247 - Val Acc: 43.45%
Epoch [03/15] | Train Loss: 0.3826 - Train Acc: 87.61% | Val Loss: 0.6415 - Val Acc: 77.10%
Epoch [04/15] | Train Loss: 0.3075 - Train Acc: 90.45% | Val Loss: 0.6484 - Val Acc: 81.42%
Epoch [05/15] | Train Loss: 0.3151 - Train Acc: 90.15% | Val Loss: 0.3615 - Val Acc: 89.54%
Epoch [06/15] | Train Loss: 0.2371 - Train Acc: 92.53% | Val Loss: 0.3389 - Val Acc: 88.88%
Epoch [07/15] | Train Loss: 0.2320 - Train Acc: 92.78% | Val Loss: 0.2374 - Val Acc: 91.29%
Epoch [08/15] | Train Loss: 0.1977 - Train Acc: 93.70% | Val Loss: 2.8916 - Val Acc: 58.30%
Epoch [09/15] | Train Loss: 0.2279 - Train Acc: 93.29% | Val Loss: 0.3191 - Val Acc: 90.20%
Epoch [10/15] | Train Loss: 0.1961 - Train Acc: 93.86% | Val Loss: 0.2495 - Val Acc: 91.59%
Epoch [11/15] | Train Loss: 0.1575 - Train Acc: 95.00% | Val Loss: 0.2433 - Val Acc: 92.03%
Epoch [12/15] | Train Loss: 0.1503 - Train Acc: 95.70% | Val Loss: 0.2767 - Val Acc: 91.59%
Epoch [13/15] | Train Loss: 0.1414 - Train Acc: 95.75% | Val Loss: 0.3002 - Val Acc: 91.59%
Epoch [14/15] | Train Loss: 0.1570 - Train Acc: 95.07% | Val Loss: 0.2720 - Val Acc: 91.51%
Epoch [15/15] | Train Loss: 0.1256 - Train Acc: 96.28% | Val Loss: 0.3190 - Val Acc: 90.34%
Best checkpoint saved: best_model/cells_VGG16.pth (epoch=11, val_acc=92.03%)
Training Complete!
```

Final Test Results (VGG16 on cells): Accuracy = 93.77% | Precision (macro) = 0.9345 | Recall (macro) = 0.9391 | Macro F1 = 0.9343

3. cells – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 8
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.4371 - Train Acc: 84.91% | Val Loss: 4.5593 - Val Acc: 40.89%
Epoch [02/15] | Train Loss: 0.2021 - Train Acc: 92.98% | Val Loss: 1.2940 - Val Acc: 69.50%
Epoch [03/15] | Train Loss: 0.1600 - Train Acc: 94.68% | Val Loss: 0.2264 - Val Acc: 91.95%
Epoch [04/15] | Train Loss: 0.1405 - Train Acc: 95.20% | Val Loss: 4.0176 - Val Acc: 46.60%
Epoch [05/15] | Train Loss: 0.1087 - Train Acc: 96.29% | Val Loss: 0.6012 - Val Acc: 85.08%
Epoch [06/15] | Train Loss: 0.1052 - Train Acc: 96.26% | Val Loss: 0.6156 - Val Acc: 84.42%
Epoch [07/15] | Train Loss: 0.1010 - Train Acc: 96.49% | Val Loss: 0.2543 - Val Acc: 91.88%
Epoch [08/15] | Train Loss: 0.0824 - Train Acc: 97.35% | Val Loss: 0.1702 - Val Acc: 94.29%
Epoch [09/15] | Train Loss: 0.0855 - Train Acc: 97.03% | Val Loss: 0.3220 - Val Acc: 89.17%
Epoch [10/15] | Train Loss: 0.0810 - Train Acc: 97.29% | Val Loss: 0.2210 - Val Acc: 92.83%
Epoch [11/15] | Train Loss: 0.0722 - Train Acc: 97.44% | Val Loss: 0.2033 - Val Acc: 93.34%
Epoch [12/15] | Train Loss: 0.0672 - Train Acc: 97.68% | Val Loss: 0.1603 - Val Acc: 94.66%
Epoch [13/15] | Train Loss: 0.0597 - Train Acc: 97.89% | Val Loss: 0.3728 - Val Acc: 90.34%
Epoch [14/15] | Train Loss: 0.0745 - Train Acc: 97.38% | Val Loss: 0.4086 - Val Acc: 89.39%
Epoch [15/15] | Train Loss: 0.0526 - Train Acc: 98.14% | Val Loss: 0.1600 - Val Acc: 95.17%
Best checkpoint saved: best_model/cells_ResNet18.pth (epoch=15, val_acc=95.17%)
Training Complete!
```

Final Test Results (ResNet18 on cells): Accuracy = 96.40% | Precision (macro) = 0.9600 | Recall (macro) = 0.9607 | Macro F1 = 0.9593

Dataset Used – chest

Chest X-ray (2 classes, grayscale) | Expected minimum test accuracy: 87%

4. chest – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
```

CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.2922	-	Train Acc: 87.22%		Val Loss: 0.2079	-	Val Acc: 93.50%
Epoch [02/15]		Train Loss: 0.1692	-	Train Acc: 93.86%		Val Loss: 0.1543	-	Val Acc: 95.03%
Epoch [03/15]		Train Loss: 0.1259	-	Train Acc: 95.09%		Val Loss: 0.1271	-	Val Acc: 94.26%
Epoch [04/15]		Train Loss: 0.1120	-	Train Acc: 95.69%		Val Loss: 0.1877	-	Val Acc: 93.12%
Epoch [05/15]		Train Loss: 0.0857	-	Train Acc: 96.96%		Val Loss: 1.4277	-	Val Acc: 78.20%
Epoch [06/15]		Train Loss: 0.1022	-	Train Acc: 96.20%		Val Loss: 0.1776	-	Val Acc: 92.93%
Epoch [07/15]		Train Loss: 0.1134	-	Train Acc: 95.94%		Val Loss: 0.1604	-	Val Acc: 96.18%
Epoch [08/15]		Train Loss: 0.0715	-	Train Acc: 97.54%		Val Loss: 0.1178	-	Val Acc: 96.18%
Epoch [09/15]		Train Loss: 0.0662	-	Train Acc: 97.56%		Val Loss: 0.0961	-	Val Acc: 96.75%
Epoch [10/15]		Train Loss: 0.0624	-	Train Acc: 97.52%		Val Loss: 0.5122	-	Val Acc: 84.32%
Epoch [11/15]		Train Loss: 0.0576	-	Train Acc: 97.86%		Val Loss: 0.0883	-	Val Acc: 96.56%
Epoch [12/15]		Train Loss: 0.0485	-	Train Acc: 98.26%		Val Loss: 0.7773	-	Val Acc: 57.55%
Epoch [13/15]		Train Loss: 0.0538	-	Train Acc: 98.26%		Val Loss: 0.1241	-	Val Acc: 96.37%
Epoch [14/15]		Train Loss: 0.0404	-	Train Acc: 98.43%		Val Loss: 0.0751	-	Val Acc: 96.94%
Epoch [15/15]		Train Loss: 0.0381	-	Train Acc: 98.45%		Val Loss: 0.1248	-	Val Acc: 96.56%

Best checkpoint saved: best_model/chest_AlexNet.pth (epoch=14, val_acc=96.94%)
Training Complete!

Final Test Results (AlexNet on chest): Accuracy = 86.22% | Precision (macro) = 0.9020 | Recall (macro) = 0.8188 | Macro F1 = 0.8392

5. chest – VGG16

Total epochs trained: 15

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.2291	-	Train Acc: 90.87%		Val Loss: 0.3031	-	Val Acc: 89.87%
Epoch [02/15]		Train Loss: 0.1500	-	Train Acc: 94.58%		Val Loss: 1.0725	-	Val Acc: 74.19%
Epoch [03/15]		Train Loss: 0.1085	-	Train Acc: 96.37%		Val Loss: 0.2111	-	Val Acc: 92.54%
Epoch [04/15]		Train Loss: 0.0874	-	Train Acc: 96.43%		Val Loss: 0.0985	-	Val Acc: 96.75%
Epoch [05/15]		Train Loss: 0.0752	-	Train Acc: 97.20%		Val Loss: 0.0941	-	Val Acc: 96.18%
Epoch [06/15]		Train Loss: 0.0873	-	Train Acc: 96.90%		Val Loss: 0.1178	-	Val Acc: 95.79%
Epoch [07/15]		Train Loss: 0.0747	-	Train Acc: 97.64%		Val Loss: 0.1277	-	Val Acc: 95.98%
Epoch [08/15]		Train Loss: 0.0679	-	Train Acc: 97.60%		Val Loss: 0.0668	-	Val Acc: 96.94%
Epoch [09/15]		Train Loss: 0.0461	-	Train Acc: 98.17%		Val Loss: 0.2930	-	Val Acc: 95.41%
Epoch [10/15]		Train Loss: 0.0649	-	Train Acc: 97.86%		Val Loss: 0.1142	-	Val Acc: 95.60%
Epoch [11/15]		Train Loss: 0.0596	-	Train Acc: 98.19%		Val Loss: 0.1478	-	Val Acc: 96.75%
Epoch [12/15]		Train Loss: 0.0508	-	Train Acc: 97.92%		Val Loss: 0.3923	-	Val Acc: 93.50%
Epoch [13/15]		Train Loss: 0.0469	-	Train Acc: 98.19%		Val Loss: 0.1677	-	Val Acc: 96.18%
Epoch [14/15]		Train Loss: 0.0359	-	Train Acc: 98.92%		Val Loss: 0.0855	-	Val Acc: 96.94%
Epoch [15/15]		Train Loss: 0.0417	-	Train Acc: 98.77%		Val Loss: 0.0937	-	Val Acc: 96.18%

Best checkpoint saved: best_model/chest_VGG16.pth (epoch=8, val_acc=96.94%)
Training Complete!

Final Test Results (VGG16 on chest): Accuracy = 85.10% | Precision (macro) = 0.8981 | Recall (macro) = 0.8030 | Macro F1 = 0.8238

6. chest – ResNet18

Total epochs trained: 15

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 2
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/15]		Train Loss: 0.2449	-	Train Acc: 89.76%		Val Loss: 0.2375	-	Val Acc: 92.54%
Epoch [02/15]		Train Loss: 0.1513	-	Train Acc: 94.22%		Val Loss: 0.2761	-	Val Acc: 90.63%
Epoch [03/15]		Train Loss: 0.1189	-	Train Acc: 95.41%		Val Loss: 0.1605	-	Val Acc: 95.22%
Epoch [04/15]		Train Loss: 0.1111	-	Train Acc: 95.77%		Val Loss: 2.8444	-	Val Acc: 44.74%
Epoch [05/15]		Train Loss: 0.0944	-	Train Acc: 96.50%		Val Loss: 0.2147	-	Val Acc: 94.07%
Epoch [06/15]		Train Loss: 0.0819	-	Train Acc: 97.13%		Val Loss: 0.3814	-	Val Acc: 84.70%
Epoch [07/15]		Train Loss: 0.0779	-	Train Acc: 96.98%		Val Loss: 0.1068	-	Val Acc: 95.22%
Epoch [08/15]		Train Loss: 0.0628	-	Train Acc: 97.56%		Val Loss: 0.1117	-	Val Acc: 95.41%
Epoch [09/15]		Train Loss: 0.0657	-	Train Acc: 97.45%		Val Loss: 0.0931	-	Val Acc: 96.56%
Epoch [10/15]		Train Loss: 0.0619	-	Train Acc: 97.62%		Val Loss: 0.1118	-	Val Acc: 95.98%
Epoch [11/15]		Train Loss: 0.0641	-	Train Acc: 97.62%		Val Loss: 0.0842	-	Val Acc: 96.56%
Epoch [12/15]		Train Loss: 0.0580	-	Train Acc: 97.94%		Val Loss: 0.1663	-	Val Acc: 95.22%
Epoch [13/15]		Train Loss: 0.0547	-	Train Acc: 97.90%		Val Loss: 0.0882	-	Val Acc: 97.32%
Epoch [14/15]		Train Loss: 0.0464	-	Train Acc: 98.41%		Val Loss: 0.2520	-	Val Acc: 94.65%
Epoch [15/15]		Train Loss: 0.0436	-	Train Acc: 98.45%		Val Loss: 0.6975	-	Val Acc: 80.69%

Best checkpoint saved: best_model/chest_ResNet18.pth (epoch=13, val_acc=97.32%)
Training Complete!

Final Test Results (ResNet18 on chest): Accuracy = 83.33% | Precision (macro) = 0.8915 | Recall (macro) = 0.7786 | Macro F1 = 0.7990

Dataset Used – lesions

Skin lesion dermatoscopy (7 classes, RGB) | Expected minimum test accuracy: 67%

7. lesions – AlexNet

Total epochs trained: 25

Configuration Used:

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

Logs:

Epoch [01/25]		Train Loss: 1.0660	-	Train Acc: 66.46%		Val Loss: 0.8400	-	Val Acc: 72.28%
Epoch [02/25]		Train Loss: 0.8856	-	Train Acc: 68.19%		Val Loss: 1.2129	-	Val Acc: 58.93%
Epoch [03/25]		Train Loss: 0.8363	-	Train Acc: 70.07%		Val Loss: 0.7677	-	Val Acc: 73.41%
Epoch [04/25]		Train Loss: 0.8196	-	Train Acc: 70.36%		Val Loss: 1.0278	-	Val Acc: 62.92%
Epoch [05/25]		Train Loss: 0.8076	-	Train Acc: 70.37%		Val Loss: 0.6878	-	Val Acc: 75.28%
Epoch [06/25]		Train Loss: 0.7711	-	Train Acc: 72.26%		Val Loss: 0.9481	-	Val Acc: 71.79%
Epoch [07/25]		Train Loss: 0.7364	-	Train Acc: 73.23%		Val Loss: 0.6866	-	Val Acc: 75.41%
Epoch [08/25]		Train Loss: 0.7343	-	Train Acc: 73.62%		Val Loss: 0.7267	-	Val Acc: 72.78%
Epoch [09/25]		Train Loss: 0.7153	-	Train Acc: 73.51%		Val Loss: 0.8142	-	Val Acc: 73.91%
Epoch [10/25]		Train Loss: 0.7011	-	Train Acc: 74.57%		Val Loss: 0.8662	-	Val Acc: 61.30%
Epoch [11/25]		Train Loss: 0.6963	-	Train Acc: 74.96%		Val Loss: 0.7687	-	Val Acc: 73.28%
Epoch [12/25]		Train Loss: 0.6717	-	Train Acc: 75.70%		Val Loss: 0.7714	-	Val Acc: 74.16%
Epoch [13/25]		Train Loss: 0.6617	-	Train Acc: 75.71%		Val Loss: 0.9457	-	Val Acc: 68.79%
Epoch [14/25]		Train Loss: 0.6486	-	Train Acc: 76.02%		Val Loss: 0.9731	-	Val Acc: 64.79%

Epoch [15/25]	Train Loss: 0.6405	- Train Acc: 76.06%	Val Loss: 0.6337	- Val Acc: 76.28%
Epoch [16/25]	Train Loss: 0.6150	- Train Acc: 77.63%	Val Loss: 0.6568	- Val Acc: 75.16%
Epoch [17/25]	Train Loss: 0.6073	- Train Acc: 77.68%	Val Loss: 0.7694	- Val Acc: 73.03%
Epoch [18/25]	Train Loss: 0.5988	- Train Acc: 78.17%	Val Loss: 0.7676	- Val Acc: 72.91%
Epoch [19/25]	Train Loss: 0.5920	- Train Acc: 78.57%	Val Loss: 0.7180	- Val Acc: 74.16%
Epoch [20/25]	Train Loss: 0.5881	- Train Acc: 78.37%	Val Loss: 1.1209	- Val Acc: 51.31%
Epoch [21/25]	Train Loss: 0.5763	- Train Acc: 78.97%	Val Loss: 0.5995	- Val Acc: 78.53%
Epoch [22/25]	Train Loss: 0.5529	- Train Acc: 80.11%	Val Loss: 0.9034	- Val Acc: 67.17%
Epoch [23/25]	Train Loss: 0.5304	- Train Acc: 80.45%	Val Loss: 0.6966	- Val Acc: 72.78%
Epoch [24/25]	Train Loss: 0.5183	- Train Acc: 81.32%	Val Loss: 0.8036	- Val Acc: 69.04%
Epoch [25/25]	Train Loss: 0.5092	- Train Acc: 81.12%	Val Loss: 0.6252	- Val Acc: 76.40%

Best checkpoint saved: best_model/lesions_AlexNet.pth (epoch=21, val_acc=78.53%)
Training Complete!

Final Test Results (AlexNet on lesions): Accuracy = 75.76% | Precision (macro) = 0.5266 | Recall (macro) = 0.4468 | Macro F1 = 0.4416

8. lesions – VGG16

Total epochs trained: 25

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4

```

Logs:

Epoch [01/25]	Train Loss: 1.0670	- Train Acc: 65.71%	Val Loss: 0.8668	- Val Acc: 70.16%
Epoch [02/25]	Train Loss: 0.9596	- Train Acc: 66.25%	Val Loss: 0.9357	- Val Acc: 70.16%
Epoch [03/25]	Train Loss: 0.9357	- Train Acc: 66.90%	Val Loss: 0.8219	- Val Acc: 70.16%
Epoch [04/25]	Train Loss: 0.9115	- Train Acc: 67.07%	Val Loss: 0.7631	- Val Acc: 71.79%
Epoch [05/25]	Train Loss: 0.9047	- Train Acc: 66.75%	Val Loss: 0.7702	- Val Acc: 70.04%
Epoch [06/25]	Train Loss: 0.8912	- Train Acc: 67.54%	Val Loss: 0.8522	- Val Acc: 69.16%
Epoch [07/25]	Train Loss: 0.9002	- Train Acc: 67.25%	Val Loss: 0.7804	- Val Acc: 72.16%
Epoch [08/25]	Train Loss: 0.8743	- Train Acc: 67.75%	Val Loss: 0.8239	- Val Acc: 71.29%
Epoch [09/25]	Train Loss: 0.8806	- Train Acc: 67.75%	Val Loss: 0.8721	- Val Acc: 71.41%
Epoch [10/25]	Train Loss: 0.8696	- Train Acc: 68.21%	Val Loss: 0.9376	- Val Acc: 73.28%
Epoch [11/25]	Train Loss: 0.8584	- Train Acc: 68.62%	Val Loss: 0.7930	- Val Acc: 72.78%
Epoch [12/25]	Train Loss: 0.8520	- Train Acc: 69.02%	Val Loss: 0.7487	- Val Acc: 72.53%
Epoch [13/25]	Train Loss: 0.8424	- Train Acc: 68.87%	Val Loss: 0.8270	- Val Acc: 72.03%
Epoch [14/25]	Train Loss: 0.8296	- Train Acc: 69.70%	Val Loss: 0.8597	- Val Acc: 62.80%
Epoch [15/25]	Train Loss: 0.8214	- Train Acc: 69.91%	Val Loss: 0.7221	- Val Acc: 74.03%
Epoch [16/25]	Train Loss: 0.8201	- Train Acc: 70.09%	Val Loss: 0.7282	- Val Acc: 73.78%
Epoch [17/25]	Train Loss: 0.8197	- Train Acc: 69.84%	Val Loss: 0.7574	- Val Acc: 71.79%
Epoch [18/25]	Train Loss: 0.8073	- Train Acc: 70.40%	Val Loss: 0.8061	- Val Acc: 70.66%
Epoch [19/25]	Train Loss: 0.8095	- Train Acc: 70.04%	Val Loss: 0.7376	- Val Acc: 73.03%
Epoch [20/25]	Train Loss: 0.8059	- Train Acc: 70.86%	Val Loss: 0.6884	- Val Acc: 74.41%
Epoch [21/25]	Train Loss: 0.8012	- Train Acc: 70.95%	Val Loss: 0.7678	- Val Acc: 72.03%
Epoch [22/25]	Train Loss: 0.7950	- Train Acc: 71.12%	Val Loss: 0.7744	- Val Acc: 72.91%
Epoch [23/25]	Train Loss: 0.7783	- Train Acc: 71.29%	Val Loss: 0.8385	- Val Acc: 68.91%
Epoch [24/25]	Train Loss: 0.8021	- Train Acc: 70.91%	Val Loss: 0.6987	- Val Acc: 75.53%
Epoch [25/25]	Train Loss: 0.7774	- Train Acc: 71.23%	Val Loss: 0.7214	- Val Acc: 71.79%

Best checkpoint saved: best_model/lesions_VGG16.pth (epoch=24, val_acc=75.53%)
Training Complete!

[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).

Final Test Results (VGG16 on lesions): Accuracy = 71.07% | Precision (macro) = 0.2646 | Recall (macro) = 0.2737 | Macro F1 = 0.2529

9. lesions – ResNet18

Total epochs trained: 25

Configuration Used:

```

ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 3
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data

```

```
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 25
LEARNING_RATE..... 0.001
NUM_CLASSES..... 7
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 0.9732 - Train Acc: 65.75% | Val Loss: 0.8721 - Val Acc: 70.16%
Epoch [02/25] | Train Loss: 0.8563 - Train Acc: 68.21% | Val Loss: 0.8612 - Val Acc: 68.91%
Epoch [03/25] | Train Loss: 0.8332 - Train Acc: 69.26% | Val Loss: 0.7671 - Val Acc: 71.79%
Epoch [04/25] | Train Loss: 0.8151 - Train Acc: 70.25% | Val Loss: 0.7813 - Val Acc: 69.04%
Epoch [05/25] | Train Loss: 0.7801 - Train Acc: 70.58% | Val Loss: 0.8015 - Val Acc: 70.16%
Epoch [06/25] | Train Loss: 0.7662 - Train Acc: 71.67% | Val Loss: 0.7538 - Val Acc: 72.28%
Epoch [07/25] | Train Loss: 0.7497 - Train Acc: 72.15% | Val Loss: 0.6827 - Val Acc: 74.28%
Epoch [08/25] | Train Loss: 0.7255 - Train Acc: 73.39% | Val Loss: 0.7255 - Val Acc: 74.78%
Epoch [09/25] | Train Loss: 0.7287 - Train Acc: 72.88% | Val Loss: 0.6965 - Val Acc: 72.91%
Epoch [10/25] | Train Loss: 0.7015 - Train Acc: 74.13% | Val Loss: 1.2668 - Val Acc: 55.93%
Epoch [11/25] | Train Loss: 0.6918 - Train Acc: 74.43% | Val Loss: 0.7917 - Val Acc: 69.41%
Epoch [12/25] | Train Loss: 0.6792 - Train Acc: 75.24% | Val Loss: 0.9410 - Val Acc: 64.42%
Epoch [13/25] | Train Loss: 0.6734 - Train Acc: 74.98% | Val Loss: 0.6490 - Val Acc: 76.40%
Epoch [14/25] | Train Loss: 0.6670 - Train Acc: 75.46% | Val Loss: 0.6657 - Val Acc: 75.41%
Epoch [15/25] | Train Loss: 0.6610 - Train Acc: 75.35% | Val Loss: 0.6653 - Val Acc: 75.16%
Epoch [16/25] | Train Loss: 0.6611 - Train Acc: 75.59% | Val Loss: 0.7217 - Val Acc: 75.16%
Epoch [17/25] | Train Loss: 0.6486 - Train Acc: 75.81% | Val Loss: 0.7617 - Val Acc: 72.16%
Epoch [18/25] | Train Loss: 0.6332 - Train Acc: 76.18% | Val Loss: 0.6605 - Val Acc: 74.91%
Epoch [19/25] | Train Loss: 0.6286 - Train Acc: 76.96% | Val Loss: 0.6927 - Val Acc: 74.16%
Epoch [20/25] | Train Loss: 0.6325 - Train Acc: 76.02% | Val Loss: 0.6595 - Val Acc: 75.53%
Epoch [21/25] | Train Loss: 0.6271 - Train Acc: 76.85% | Val Loss: 0.7568 - Val Acc: 74.16%
Epoch [22/25] | Train Loss: 0.6124 - Train Acc: 77.33% | Val Loss: 0.6221 - Val Acc: 76.65%
Epoch [23/25] | Train Loss: 0.5968 - Train Acc: 77.78% | Val Loss: 0.6373 - Val Acc: 78.40%
Epoch [24/25] | Train Loss: 0.6004 - Train Acc: 77.39% | Val Loss: 0.6339 - Val Acc: 76.78%
Epoch [25/25] | Train Loss: 0.5895 - Train Acc: 78.10% | Val Loss: 0.6557 - Val Acc: 75.16%
Best checkpoint saved: best_model/lesions_ResNet18.pth (epoch=23, val_acc=78.40%)
Training Complete!
```

[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).

Final Test Results (ResNet18 on lesions): Accuracy = 75.21% | Precision (macro) = 0.5463 | Recall (macro) = 0.4072 | Macro F1 = 0.4354

Dataset Used – orgs

Organ images (11 classes, grayscale) | Expected minimum test accuracy: 83%

10. orgs – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.9865 - Train Acc: 65.32% | Val Loss: 0.3918 - Val Acc: 86.20%
Epoch [02/15] | Train Loss: 0.3739 - Train Acc: 87.52% | Val Loss: 0.8075 - Val Acc: 76.89%
Epoch [03/15] | Train Loss: 0.2819 - Train Acc: 90.60% | Val Loss: 0.8631 - Val Acc: 77.86%
Epoch [04/15] | Train Loss: 0.2297 - Train Acc: 92.29% | Val Loss: 0.2273 - Val Acc: 91.73%
Epoch [05/15] | Train Loss: 0.1812 - Train Acc: 93.78% | Val Loss: 0.3824 - Val Acc: 88.09%
Epoch [06/15] | Train Loss: 0.1751 - Train Acc: 94.16% | Val Loss: 0.2326 - Val Acc: 92.58%
Epoch [07/15] | Train Loss: 0.1602 - Train Acc: 94.84% | Val Loss: 0.1579 - Val Acc: 94.01%
Epoch [08/15] | Train Loss: 0.1452 - Train Acc: 95.27% | Val Loss: 0.4226 - Val Acc: 87.76%
Epoch [09/15] | Train Loss: 0.1568 - Train Acc: 94.95% | Val Loss: 0.1755 - Val Acc: 94.08%
Epoch [10/15] | Train Loss: 0.1196 - Train Acc: 95.98% | Val Loss: 0.3262 - Val Acc: 90.69%
```

```
Epoch [11/15] | Train Loss: 0.1254 - Train Acc: 95.84% | Val Loss: 0.1107 - Val Acc: 96.29%
Epoch [12/15] | Train Loss: 0.1151 - Train Acc: 96.36% | Val Loss: 1.2548 - Val Acc: 77.21%
Epoch [13/15] | Train Loss: 0.1015 - Train Acc: 96.70% | Val Loss: 0.1684 - Val Acc: 94.66%
Epoch [14/15] | Train Loss: 0.0970 - Train Acc: 96.85% | Val Loss: 0.1226 - Val Acc: 95.77%
Epoch [15/15] | Train Loss: 0.0925 - Train Acc: 97.08% | Val Loss: 0.1383 - Val Acc: 95.44%
Best checkpoint saved: best_model/orgs_AlexNet.pth (epoch=11, val_acc=96.29%)
Training Complete!
```

Final Test Results (AlexNet on orgs): Accuracy = 87.82% | Precision (macro) = 0.8745 | Recall (macro) = 0.8600 | Macro F1 = 0.8630

11. orgs – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 1.4057 - Train Acc: 48.57% | Val Loss: 1.1373 - Val Acc: 63.93%
Epoch [02/15] | Train Loss: 0.6335 - Train Acc: 76.11% | Val Loss: 0.5719 - Val Acc: 80.60%
Epoch [03/15] | Train Loss: 0.5131 - Train Acc: 81.12% | Val Loss: 0.4507 - Val Acc: 82.55%
Epoch [04/15] | Train Loss: 0.4037 - Train Acc: 84.44% | Val Loss: 0.3259 - Val Acc: 88.02%
Epoch [05/15] | Train Loss: 0.3323 - Train Acc: 87.61% | Val Loss: 0.2621 - Val Acc: 90.49%
Epoch [06/15] | Train Loss: 0.2879 - Train Acc: 88.83% | Val Loss: 0.2019 - Val Acc: 90.76%
Epoch [07/15] | Train Loss: 0.2734 - Train Acc: 89.28% | Val Loss: 0.2201 - Val Acc: 90.36%
Epoch [08/15] | Train Loss: 0.2349 - Train Acc: 90.41% | Val Loss: 0.2399 - Val Acc: 91.80%
Epoch [09/15] | Train Loss: 0.2443 - Train Acc: 90.41% | Val Loss: 0.2133 - Val Acc: 90.49%
Epoch [10/15] | Train Loss: 0.2299 - Train Acc: 91.61% | Val Loss: 0.1921 - Val Acc: 92.97%
Epoch [11/15] | Train Loss: 0.1939 - Train Acc: 93.56% | Val Loss: 0.1430 - Val Acc: 94.86%
Epoch [12/15] | Train Loss: 0.1700 - Train Acc: 94.17% | Val Loss: 0.1846 - Val Acc: 94.40%
Epoch [13/15] | Train Loss: 0.1647 - Train Acc: 94.66% | Val Loss: 0.1257 - Val Acc: 94.60%
Epoch [14/15] | Train Loss: 0.1556 - Train Acc: 95.10% | Val Loss: 0.1437 - Val Acc: 95.77%
Epoch [15/15] | Train Loss: 0.1939 - Train Acc: 94.00% | Val Loss: 0.2667 - Val Acc: 92.97%
Best checkpoint saved: best_model/orgs_VGG16.pth (epoch=14, val_acc=95.77%)
Training Complete!
```

Final Test Results (VGG16 on orgs): Accuracy = 89.69% | Precision (macro) = 0.8845 | Recall (macro) = 0.8811 | Macro F1 = 0.8815

12. orgs – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION..... ReLU
BATCH_SIZE..... 64
CHANNELS..... 1
CRITERION..... CrossEntropyLoss
DATA_PATH..... ../data
DEVICE..... cuda
DROP_RATE..... 0.5
EPOCHS..... 15
LEARNING_RATE..... 0.001
NUM_CLASSES..... 11
OPTIMIZER..... Adam
RANDOM_SEED..... 42
VAL_SPLIT..... 0.1
WEIGHT_DECAY..... 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.6892 - Train Acc: 75.27% | Val Loss: 0.5103 - Val Acc: 81.38%
Epoch [02/15] | Train Loss: 0.3209 - Train Acc: 88.52% | Val Loss: 0.2660 - Val Acc: 90.49%
Epoch [03/15] | Train Loss: 0.2403 - Train Acc: 91.67% | Val Loss: 0.2628 - Val Acc: 90.69%
Epoch [04/15] | Train Loss: 0.2114 - Train Acc: 92.58% | Val Loss: 0.1888 - Val Acc: 93.68%
Epoch [05/15] | Train Loss: 0.1760 - Train Acc: 93.72% | Val Loss: 0.1490 - Val Acc: 94.40%
Epoch [06/15] | Train Loss: 0.1347 - Train Acc: 95.23% | Val Loss: 0.1926 - Val Acc: 92.51%
```

Epoch [07/15] | Train Loss: 0.1173 - Train Acc: 95.91% | Val Loss: 0.1213 - Val Acc: 95.38%
Epoch [08/15] | Train Loss: 0.0931 - Train Acc: 96.86% | Val Loss: 0.1454 - Val Acc: 94.73%
Epoch [09/15] | Train Loss: 0.0953 - Train Acc: 96.72% | Val Loss: 0.1178 - Val Acc: 95.96%
Epoch [10/15] | Train Loss: 0.0857 - Train Acc: 96.98% | Val Loss: 0.0805 - Val Acc: 96.88%
Epoch [11/15] | Train Loss: 0.0690 - Train Acc: 97.54% | Val Loss: 0.1016 - Val Acc: 96.09%
Epoch [12/15] | Train Loss: 0.0496 - Train Acc: 98.36% | Val Loss: 0.1253 - Val Acc: 95.57%
Epoch [13/15] | Train Loss: 0.0468 - Train Acc: 98.24% | Val Loss: 0.2232 - Val Acc: 93.29%
Epoch [14/15] | Train Loss: 0.0583 - Train Acc: 97.88% | Val Loss: 0.0899 - Val Acc: 96.48%
Epoch [15/15] | Train Loss: 0.0392 - Train Acc: 98.58% | Val Loss: 0.1398 - Val Acc: 96.09%
Best checkpoint saved: best_model/orgs_ResNet18.pth (epoch=10, val_acc=96.88%)
Training Complete!

Final Test Results (ResNet18 on orgs): Accuracy = 90.63% | Precision (macro) = 0.9054 | Recall (macro) = 0.8851 | Macro F1 = 0.8923

Consolidated Summary – All Permutations

Dataset	Model	Accuracy (%)	Precision	Recall	Macro F1
cells	AlexNet	95.50	0.955	0.948	0.951
cells	VGG16	93.77	0.934	0.939	0.934
cells	ResNet18	96.40	0.960	0.961	0.959
chest	AlexNet	86.22	0.902	0.819	0.839
chest	VGG16	85.10	0.898	0.803	0.824
chest	ResNet18	83.33	0.892	0.779	0.799
lesions	AlexNet	75.76	0.527	0.447	0.442
lesions	VGG16	71.07	0.265	0.274	0.253
lesions	ResNet18	75.21	0.546	0.407	0.435
orgs	AlexNet	87.82	0.875	0.860	0.863
orgs	VGG16	89.69	0.884	0.881	0.881
orgs	ResNet18	90.63	0.905	0.885	0.892

Architectural Recommendations

Based on the Part 1 benchmark results across all twelve dataset-model permutations, the recommended pairings are:

Dataset	Recommended model	Test accuracy	Macro F1	Rationale
cells	ResNet18	96.40%	0.959	Highest accuracy and F1; AlexNet (95.50%) is a strong lighter alternative when compute is limited.
chest	AlexNet	86.22%	0.839	Best of the three on this 2-class task; VGG16 (85.10%) and ResNet18 (83.33%) trail slightly.
lesions	AlexNet	75.76%	0.442	Highest accuracy on this hard, imbalanced 7-class set; ResNet18 (75.21%) is close. Macro F1 is low for all models due to class imbalance.
orgs	ResNet18	90.63%	0.892	Best accuracy, precision, recall, and F1; residual depth pays off on this 11-class set.

Summary: No single architecture dominates: ResNet18 gives the best accuracy on the larger multi-class sets (cells and orgs), while AlexNet leads on chest and lesions. ResNet18 is the safest all-round default, and AlexNet is the most efficient pick wherever the accuracy gap is small.

Part 2 - Green Initiative: Architectural Downscaling and Efficiency Analysis

Objective

The original model registry (AlexNet, VGG16, ResNet18) relies on heavy, over-parameterized topologies. This phase redesigns them to drastically cut computational and energy cost while preserving baseline classification accuracy, and quantifies the trade-off across every dataset-model permutation.

Architectural changes

For AlexNet and VGG16, the fully-connected classifier heads (which held the vast majority of parameters) were replaced with a single global average pooling (GAP) layer followed by one linear layer; VGG16 was additionally shortened from five convolutional blocks to three. For ResNet18, each stage was reduced from two residual blocks to one. All models keep the same (in_channels, num_classes) interface and were trained and evaluated under identical settings as the baseline.

Efficiency Verification Matrix

Dataset	Model	Params (M)	Train time (s)	Inf. latency (ms/sample)	Peak mem train / infer (MB)
cells	AlexNet	5.69 to 1.49	15 to 16	0.030 to 0.051	193 to 125 / 133 to 66
cells	VGG16	12.63 to 1.22	62 to 45	0.103 to 0.082	873 to 619 / 408 to 232
cells	ResNet18	11.17 to 4.90	130 to 68	0.206 to 0.114	1336 to 828 / 515 to 351
chest	AlexNet	5.69 to 1.49	5 to 4	0.039 to 0.035	190 to 123 / 130 to 64
chest	VGG16	12.63 to 1.22	23 to 17	0.109 to 0.088	876 to 616 / 407 to 229
chest	ResNet18	11.17 to 4.90	50 to 26	0.223 to 0.120	1333 to 826 / 510 to 350
lesions	AlexNet	5.69 to 1.49	20 to 11	0.057 to 0.033	193 to 127 / 132 to 67
lesions	VGG16	12.63 to 1.22	66 to 46	0.126 to 0.099	876 to 618 / 408 to 231
lesions	ResNet18	11.17 to 4.90	128 to 71	0.211 to 0.137	1336 to 827 / 514 to 351
orgs	AlexNet	5.69 to 1.49	13 to 17	0.025 to 0.041	190 to 123 / 132 to 64
orgs	VGG16	12.63 to 1.22	68 to 52	0.098 to 0.088	876 to 616 / 407 to 229
orgs	ResNet18	11.17 to 4.90	141 to 76	0.198 to 0.110	1333 to 826 / 510 to 350

Accuracy preservation

Dataset	Model	Accuracy (%)	Macro F1
cells	AlexNet	95.5 to 94.7	0.951 to 0.943
cells	VGG16	93.8 to 96.3	0.934 to 0.965
cells	ResNet18	96.4 to 95.4	0.959 to 0.950
chest	AlexNet	86.2 to 85.9	0.839 to 0.834
chest	VGG16	85.1 to 83.0	0.824 to 0.794
chest	ResNet18	83.3 to 84.1	0.799 to 0.811
lesions	AlexNet	75.8 to 74.0	0.442 to 0.471
lesions	VGG16	71.1 to 76.2	0.253 to 0.484
lesions	ResNet18	75.2 to 75.1	0.435 to 0.464
orgs	AlexNet	87.8 to 89.4	0.863 to 0.881
orgs	VGG16	89.7 to 91.9	0.881 to 0.908
orgs	ResNet18	90.6 to 92.8	0.892 to 0.918

Accuracy stayed comparable, within roughly 1-2 points either way in most cases. The slimmed models improved across the board on orgs and lifted lesions-VGG16 sharply (+5.1), while the main change was chest-VGG16 (85.1 to 83.0). This mixed pattern indicates the smaller models retain enough capacity for these datasets without a systematic accuracy penalty.

Trade-off discussion

The slimmed architectures deliver comparable accuracy at a fraction of the computational cost. The clearest evidence is VGG16: a 90% parameter cut (12.6M to 1.2M) with accuracy holding or rising on three of four datasets, which shows the original fully-connected head was largely redundant capacity for these small 64x64 inputs. Replacing that head with global average pooling both removed parameters and added a mild regularizing effect.

Model size is not the same as real cost. ResNet18 has fewer parameters than the original VGG16 yet remains the slowest and most memory-hungry, because its depth and residual activations drive runtime and memory more than raw weight count.

Slimming did not help uniformly - chest-VGG16 lost about two points - so the smallest configurations are not always sufficient. And on lesions (a heavily imbalanced 7-class set, ~66% one class) macro F1 stays low across all models, reflecting weak performance on the rare classes.

Recommendation

The green-optimized architectures are recommended for deployment: they hold or improve accuracy on most permutations while cutting parameters by 56-90%, training runtime by up to 48%, and inference memory by 40-50%. The slimmed VGG16 is the standout, improving accuracy on most datasets at a tenth of the parameters. Future work is to keep slightly more capacity for chest, and apply class-balancing or augmentation on lesions to lift the low macro F1.